Prostředí pro vizuální návrh, monitorování a správu distribuovaných řídicích systémů a IoT sítí

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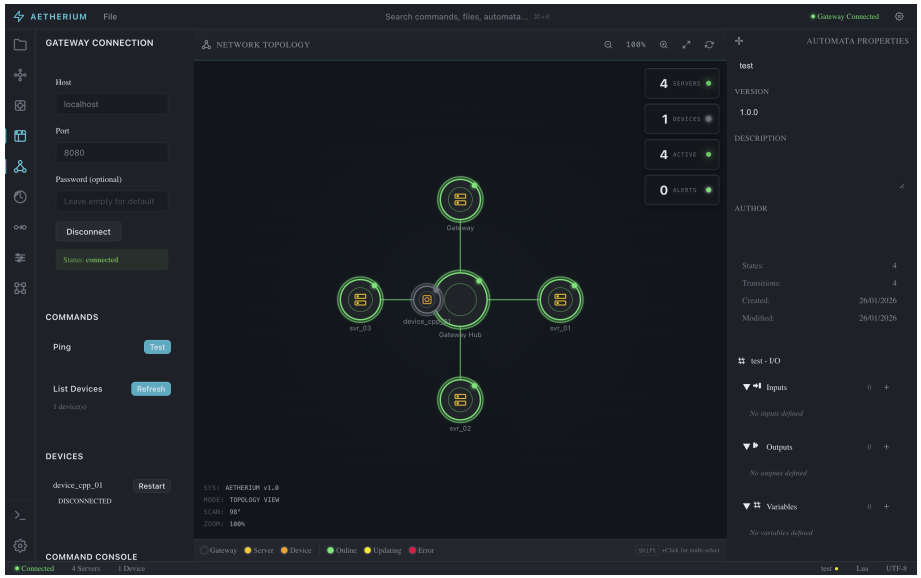
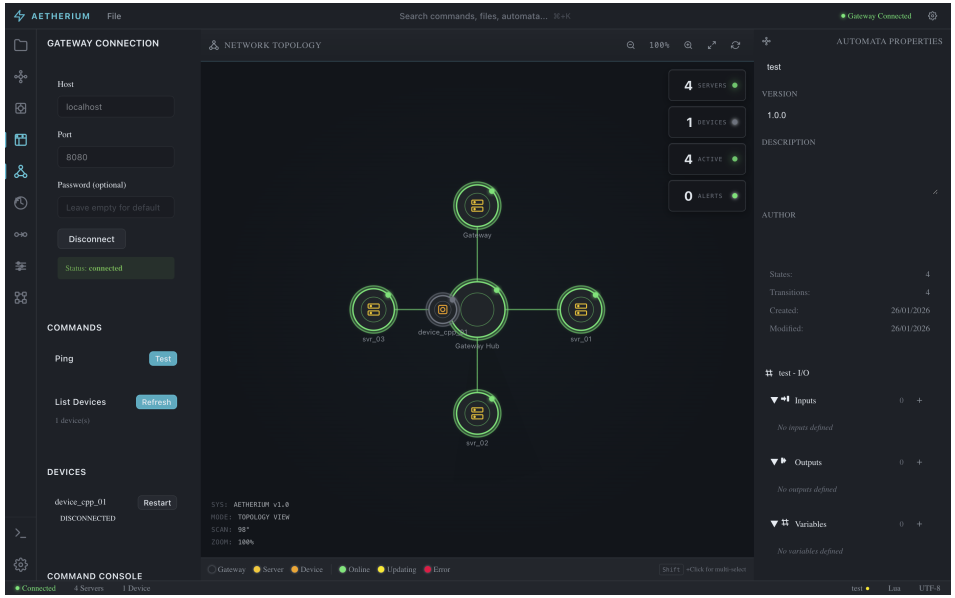
Motivation

- Distributed control systems are hard to prototype & debug.
- Goal: a lightweight runtime that executes communicating timed automata.
- Same model, multiple nodes, observable behaviour.



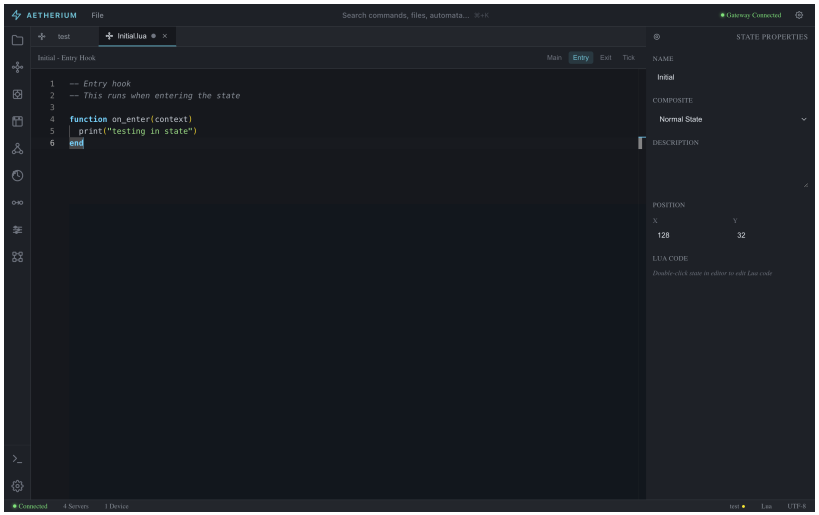
System Architecture (Current State)

- Timed automata run on multiple nodes.
- Gateway/server connect, coordinate, and observe.



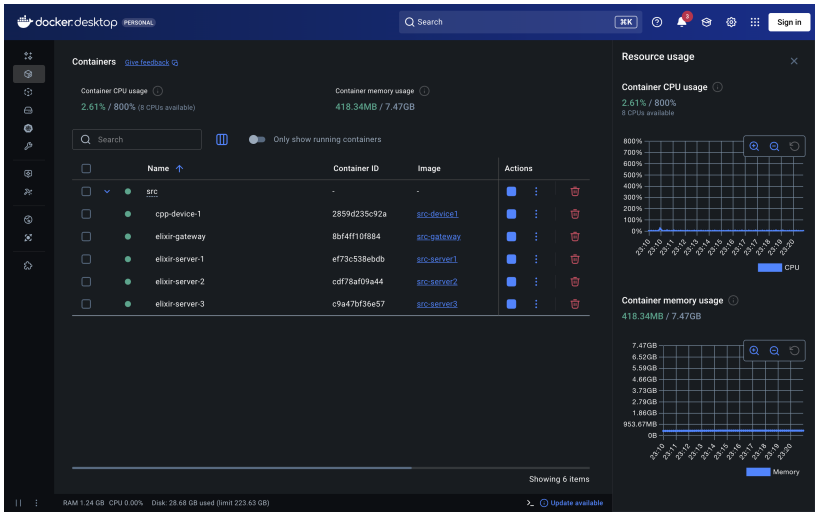
Runtime: Timed Automata + Scripting

- YAML model: states, transitions, variables.
- Lua hooks for custom logic.



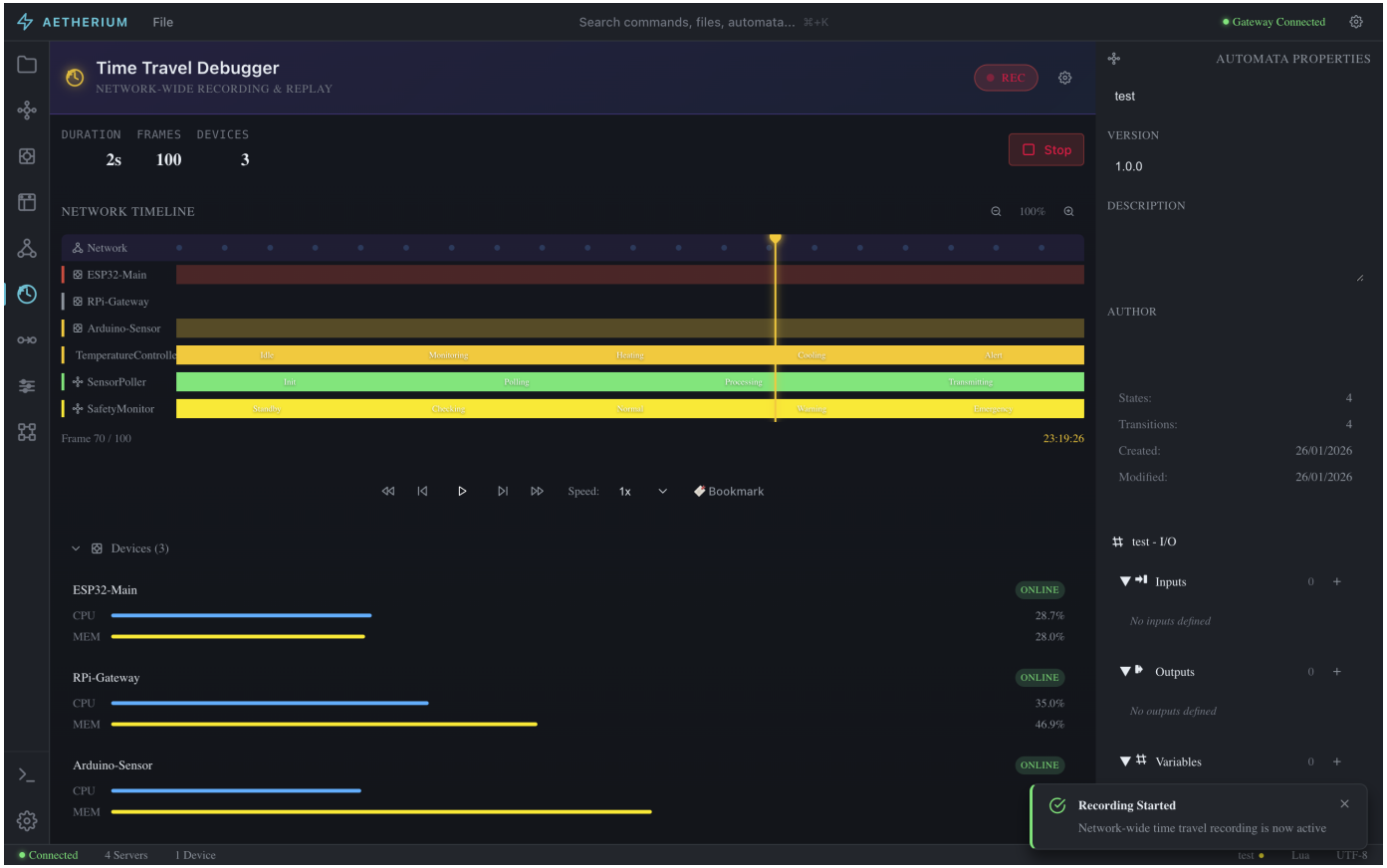
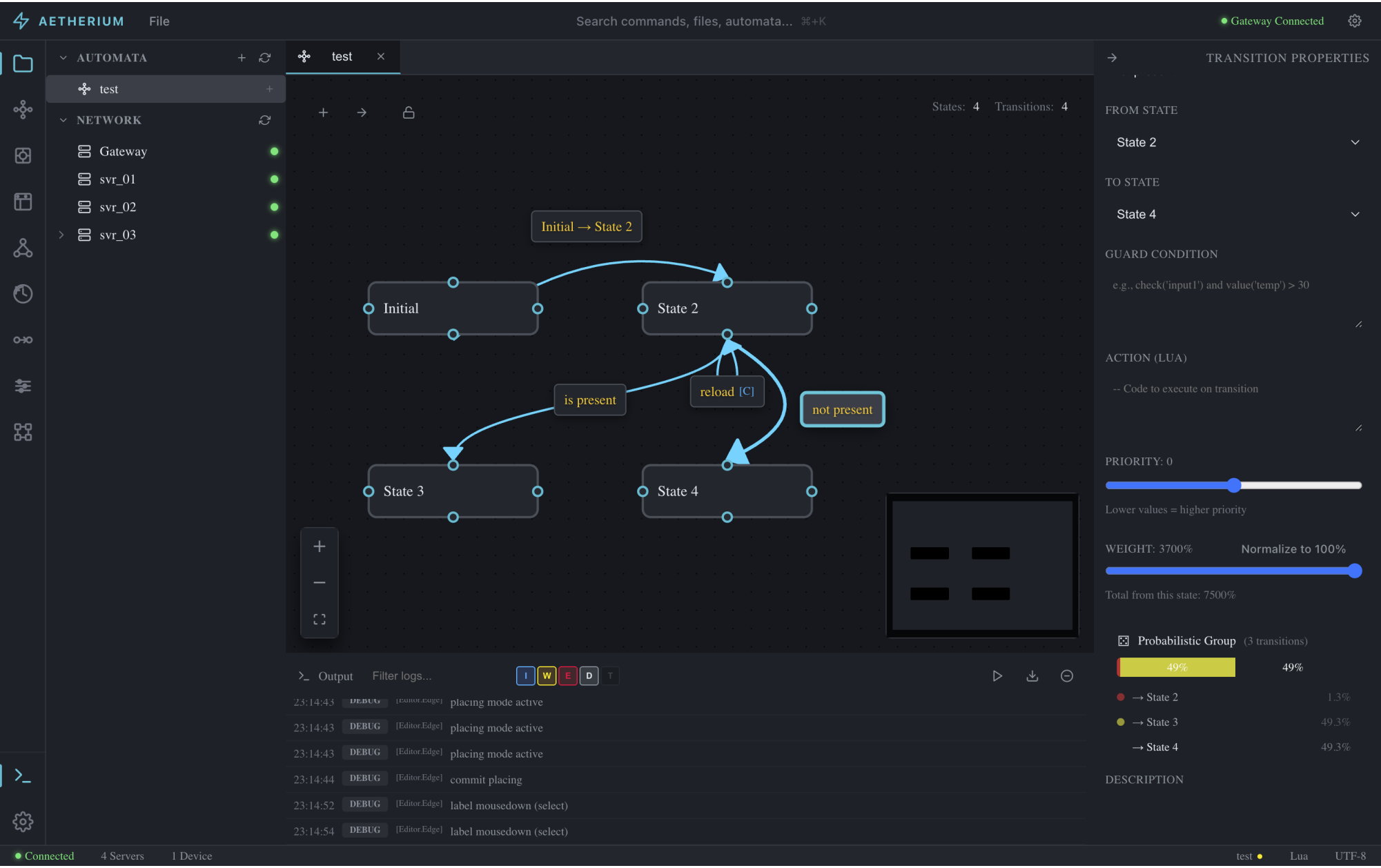
Deployment / Infrastructure (Current State)

- Docker-based services for repeatable development.
- Tested gateway/server + engine CLI.



Finished Requirements (Summary)

- **Studied:** DEVS, IEC 61499, Node-RED, 4diac/FORTE, ROS2/Micro-ROS.
- **Implemented / in progress:** runtime + protocol + services; IDE/time-travel/reconfiguration next.



Thank You for Your Attention!